

X-ray radiation effects on thin film nanocomposites of functionalized and copper coated multi-walled carbon nanotube and poly(methyl methacrylate)

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ABSTRACT

Multi-walled carbon nanotube (MWCNT) and poly(methyl methacrylate) (PMMA) nanocomposites have the potential to be deployed as sensing elements for various applications including ionizing radiation detection. In this work, a 160 kV x-ray ionizing radiation source has been used to study the ionizing radiation dose rate effects on Ferrocene doped MWCNTs, MWCNTs functionalized with - carboxylic acid (COOH), hydrogen peroxide, amine, and MWCNTs coated/ modified with copper. Solution blending method was used in synthesizing the nanocomposites and the synthesized composites were cast on interdigitated electrodes (IDEs) to perform radiation sensing experiments by monitoring changes in the films' resistance. We have observed a clear, repeatable and proportionate response curve for radiation dose rate measurements with Ferrocene doped MWCNT/PMMA thin film composites. A weaker response is observed for functionalized MWCNT/PMMA films. Ferrocene doped MWCNT/PMMA composite showed the most increased sensitivity to x-ray radiation when compared to all other functional groups reported here and as compared to unfunctionalized MWCNT/PMMA composites. Ferrocene doped MWCNT/PMMA films' resistance changed over 18% compared to changes observed for unfunctionalized MWCNT/PMMA films of about 6%. Accelerated post radiation recovery of Ferrocene doped MWCNT/PMMA composites using a heat treatment technique is presented. The results of this study suggest that simple sensor elements based on relatively economical MWCNTs could be useful for the detection of ionizing radiation.

1. Introduction

Characterizing and monitoring the radiation environment for aerospace and terrestrial applications requires sensitive, light weight, rugged and compact instrumentation. Some existing sensor types used to measure radiation [1] are problematic because they are bulky and provide non-instantaneous response characteristics [2]. Some of the exceptional properties of carbon nanotubes (CNTs), such as very high surface area, high surface to volume ratio, electrical properties, and chemical reactivity, make CNTs an excellent potential candidate material for small and robust sensors for various applications [3].

Many studies have used CNTs as active components in detecting various gases due to their low power consumption, high response, repeatability, and low temperature operation properties [4]. Conductivity change is detected in the nanotube when there is transfer of electrons between CNTs and the adsorbed gas molecule [5]. CNTs functionally modified with polymer additives [4], functional groups [4,6], and metal nanoparticles [4] have shown enhanced conductivity response

when used as sensor elements due to the increase in the electron transfer rate [7].

Several studies have been performed on carboxylic acid (COOH) [6,8–10], polyethyleneimine [11], and copper nano-particle [12] functionalized or modified CNTs for sensing chloroform, ethanol, glucose, and various gases. CNTs functionalized or modified with various groups such as COOH [13,14], copper [15], and 3-aminopropyltriethoxysilane [16] have shown increased sensitivity to alcohol vapors, glucose and various gases compared to unfunctionalized CNTs. Various types of functionalized SWCNT response to proton radiation have also been studied [2]. These studies show that the use of functionalized CNTs in sensor application studies tend to increase sensitivity with respect to unfunctionalized CNTs. These works motivated us to explore the possibility of exploiting the same property for a radiation sensing element for aerospace and other applications.

Multi-walled carbon nanotubes (MWCNTs), which are generally considered cheaper and/or easier to process than graphene or SWCNTs, are used here to study the response of a nano-scaled carbon material to

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ionizing radiation in the form of x-rays. In previous work, it was observed that unfunctionalized MWCNT/PMMA films showed a clear, repeatable and proportional response to various doses and dose rates of x-rays [17]. In the present work, a 160 kV x-ray ionizing radiation source has been used to study the radiation dose rate effects on nanocomposites made with ferrocene doped MWCNTs, MWCNTs functionalized with carboxylic acid, hydrogen peroxide, amine, and MWCNTs coated/modified with copper. The response of these nanocomposites are compared to similar nanocomposites made with unfunctionalized MWCNTs [17]. The specific functionalized and modified MWCNTs used for this study were determined by those used in previous literature and those that were readily available commercially and inexpensively. In addition, the commercially functionalized MWCNTs used here are from the same source as the unfunctionalized MWCNT, thus making a more straightforward comparison with data from unfunctionalized [17] MWCNT/PMMA thin film nanocomposites. Similar to the composites made with unfunctionalized MWCNT, all composites with the functional groups and copper coated (modified) MWCNTs exhibited smaller change in the film resistance for higher x-ray dose rates. Ferrocene doped MWCNT showed increased sensitivity to x-ray radiation compared to unfunctionalized MWCNT [17] and all other functional groups and copper-coated MWCNTs used in this study. To our knowledge, this is the first study of the electrical conductivity effects on functionalized and modified MWCNT/PMMA composites when exposed to x-rays.

2. Materials and methods

2.1. Raw materials

Unfunctionalized and functionalized/modified MWCNT (COOH, hydrogen peroxide, amine and copper coated) used in this study were commercially produced by the thermal chemical vapor deposition method with inner diameter of 10 nm, outer diameter of 30 ± 15 nm, length of 1–5 μm , total number of walls 10–15 and purity >95%, and were obtained from Nanolab, Inc. Ferrocene, an organo-metallic compound was purchased from Sigma Aldrich. To obtain uniform nanotube dispersion [18], *N,N*-Dimethylformamide (DMF) purchased from Sigma Aldrich was used as the solvent. PMMA purchased from Sigma Aldrich [$T_g = 105^\circ$] was chosen as the polymer because of its spinning qualities [19], solubility in DMF [19], high weatherability [20], and good chemical resistance [21]. Inter-digitated electrodes (IDE), supplied by Case Western Reserve University shown in Fig. 1(a), were used as the substrate to cast the synthesized thin film composite. The IDE is patterned on an 8 mm x 8 mm, 0.3 mm thick silicon substrate with a top 300 nm thick silicon dioxide layer. The electrodes and bonding pads (1.75 x 1.5 mm) on the IDE are made from 200 nm thick gold with a 10 nm thick titanium adhesion layer. The electrodes run parallel to each other with a line size and spacing of 100 μm .

2.2. Functionalized MWCNT/PMMA nanocomposite preparation

Ferrocene doped MWCNT/PMMA nanocomposites were prepared by solution blending technique [22]. Ferrocene in DMF solution with a concentration of 2.62 mg/cm^3 was dissolved and tip sonicated in the presence of an ice bath with 0.03 g/cm^3 of PMMA in DMF solution for 15 min at 20% amplitude and a pulse rate of 0.5 using a 750 W, 20 kHz GENEC probe sonicator. To the above solution, 1 mg/cm^3 of unfunctionalized MWCNT in DMF solution was mixed and sonicated in the presence of an ice bath for an additional 5 min. The sonicated mixture was then spin coated at 1000 rpm, with a ramp of 100 rpm/s for 30 s on the IDE using a Brewer Science spin coater to form a carbon nanotube network bridging the IDE fingers. The cast film was cured in a vacuum oven at 175°C for 1 h and cooled under a fume hood overnight to remove excess solvent.

The functionalized and modified MWCNT (COOH, hydrogen peroxide, amine and copper coated) were prepared in a similar manner as

ferrocene doped MWCNT/PMMA composite, but without incorporating ferrocene into the composite and with different MWCNT weight and sonication times as described in Table 1.

2.3. Experimental setup

Radiation dose rate experiments were performed using a 160 kV, 3000 W “X-RAD iR-160” x-ray source with a tungsten target (anode) at a focal spot of 7.5 mm purchased from Precision X-Ray. The x-ray source produces a broad spectra with well-defined peaks at 55.55 and 66.66 keV.

Radiation dose rate effects were studied by varying the source to sample distance (SSD) (30, 50, and 70 cm). The vendor specified dose rates of 5.92 krad/min at 30 cm SSD was used to calculate the dose rates for 50 cm SSD and 70 cm SSD using the inverse square law.

Radiation exposures were performed on the composite samples placed in the x-ray chamber and *in situ* electrical measurements were performed by connecting a low noise cable from the x-ray chamber to a Hewlett Packard HP4145B semiconductor parametric analyser. Commercial control software, Interactive Characterization Software version-3.9 on a personal computer interfaced with the HP4145B, was programmed to record the electrical measurements automatically every one minute before, during and after radiation. The experimental setup for these standard electrical measurements is described in more detail in Ref. [17].

3. Results

Radiation dose rate response of each MWCNT/PMMA composite film was studied by varying the SSD of the x-ray source described in Section 2.3. Change in resistance (S) is used as a parameter in detecting the response of the composite films to ionizing radiation in this work, which is shown in Eq. (1):

$$S = \left(\frac{R_f - R_i}{R_i} \right) \times 100\% \quad (1)$$

where R_f is the final resistance after radiation exposure to a particular fixed dose with varying dose rates at a given time, as measured each minute by the parametric analyzer, and R_i is the initial resistance before radiation exposure.

3.1. Radiation dose rate response

Radiation dose rate effects were studied by placing the fabricated MWCNT/PMMA nanocomposite thin film samples at the center of the 160 kV x-ray source and by varying their distance from the x-ray source. Dose rates of 5.92, 2.13, and 1.08 krad/min were achieved by placing the samples at 30, 50, and 70 cm from the x-ray source respectively with a constant total dose of 100 krad for each dose rate experiments. Electrical resistance of the fabricated thin films under test, and a nominally identical control sample not exposed to radiation, were sampled at 1 min intervals; before, during and after radiation exposure.

Results of the dose rate experiments for Ferrocene doped (Fig. 2) MWCNT/PMMA nanocomposite thin film clearly exhibits a continuous decrease in electrical resistance of the composite during radiation exposure with a distinctive response curve. The magnitude of the reduction in electrical resistance of the composite is roughly inversely proportional to the radiation dose rate. The control sample did not show any significant change during the time of measurements.

The Carboxylic acid, copper coated, amine and hydrogen peroxide functionalized MWCNT/PMMA nanocomposites show a similar response, but the magnitude of the resistance change is significantly less as compared to the ferrocene doped MWCNT/PMMA nanocomposites for all dose rates.

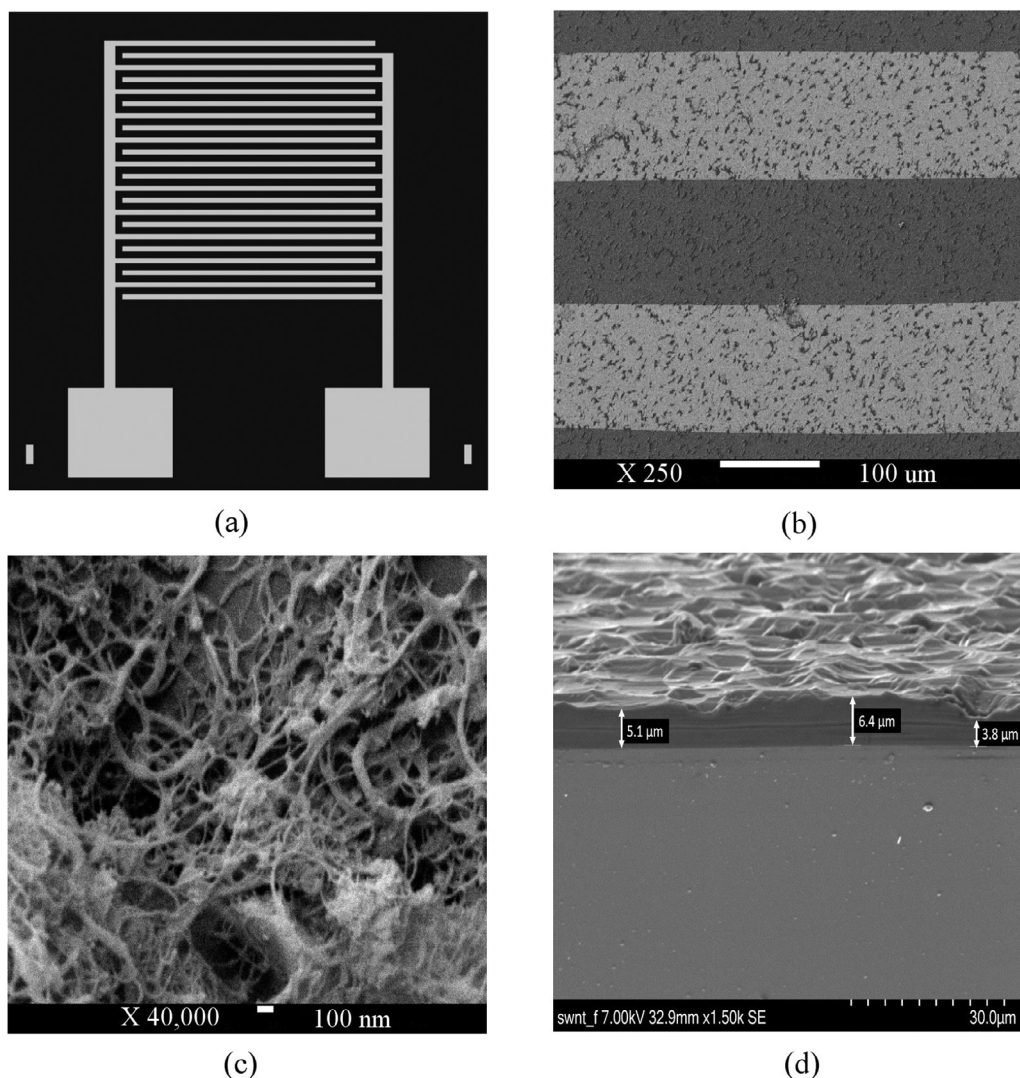


Fig. 1. (a) Graphical representation of an IDE. (b), (c) SEM images of dispersed COOH functionalized MWCNT/PMMA composite on an IDE. (d) SEM image of ferrocene functionalized MWCNT/PMMA composite showing the film thickness ranging from 3 μ to 7 μ .

Table 1
Weight and sonication time of functionalized MWCNT.

Functionalized MWCNT	Weight in DMF solution	Sonication time
Carboxylic acid	5 mg/cm ³	7 min
Amine	2 mg/cm ³	5 min
Hydrogen peroxide	1 mg/cm ³	5 min
Copper coated	0.6 mg/cm ³	8 min

3.2. X-ray radiation sensitivity comparison of functionalized and unfunctionalized MWCNT/PMMA composite

Radiation dose rate experimental results for functionalized (ferrocene doped, Carboxylic acid, amine, hydrogen peroxide and copper coated) and unfunctionalized [17] MWCNT/PMMA thin film composites are summarized in Fig. 3 for three different dose rates.

As seen from the graph in Fig. 3, ferrocene doped MWCNT/PMMA nanocomposites outperforms unfunctionalized MWCNT/PMMA composites [17] and all other functional groups used in this study with respect to resistance change upon x-ray exposure across all radiation dose rates.

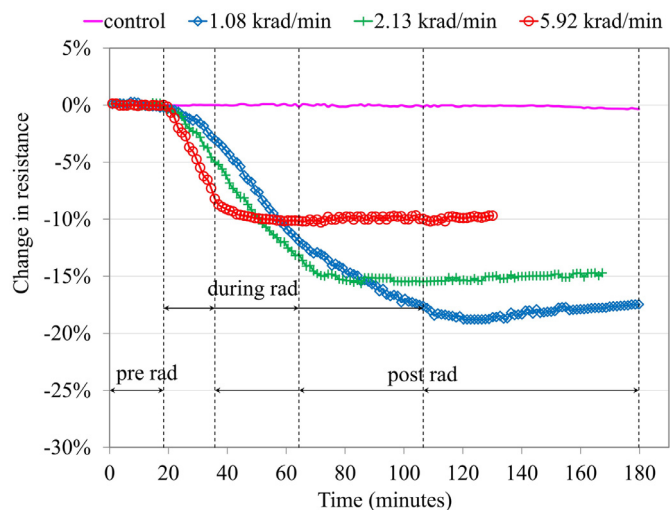


Fig. 2. Change in resistance over time for Ferrocene doped MWCNT/PMMA composite for varying dose rates (total dose in each case was 100 krad).

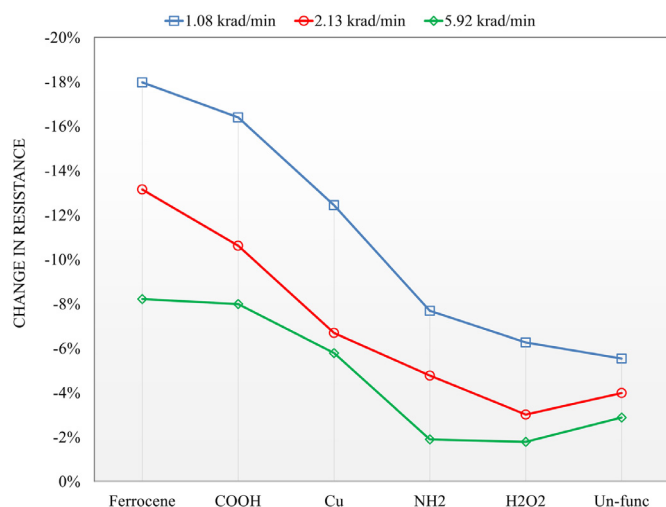


Fig. 3. Change in resistance with varying dose rates for unfunctionalized and functionalized MWCNT/PMMA composite. (Total dose in each case was 100 krad.).

3.3. Repeatability and stability of the thin film composite

The natural recovery time for ferrocene doped MWCNT/PMMA thin film samples after irradiation was on the order of one week, with similar times for the other nanocomposites. The results showing repeatability of the dose rate experiments for ferrocene doped MWCNT/PMMA composites are shown in Fig. 4. This first cycle from un-irradiated sample to the recovered state is called Trial-1. The x-ray response experiments were repeated on the recovered samples to check for repeatability of their response, this was Trial-2. The results clearly indicate that the Ferrocene doped MWCNT/PMMA composite thin films have a repeatable proportionate electrical resistance change response to varying radiation dose rates. Similar results were obtained for the Carboxylic acid, copper coated, amine and hydrogen peroxide functionalized MWCNT/PMMA composite films. For the ferrocene doped MWCNT/PMMA composites, the differences measured between the trials was on average less than 10.5%. The fairly small changes observed between trials for the radiation dose rate measurements are indicative of the stability of the functionalized MWCNT/PMMA thin films during the measurements presented.

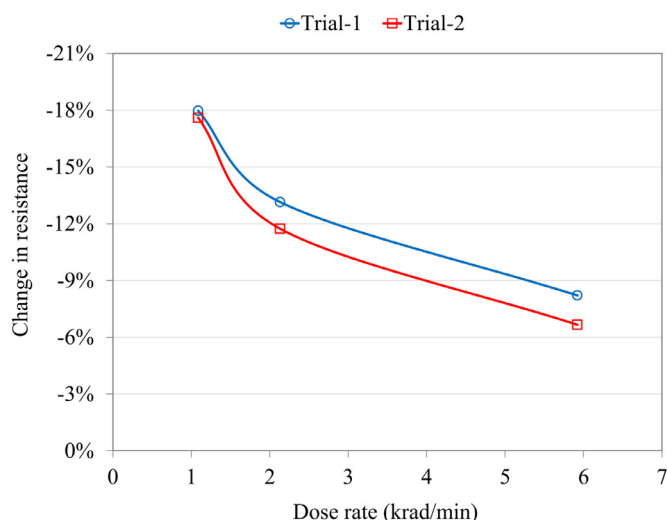


Fig. 4. Change in resistance with varying dose rate for Ferrocene doped MWCNT/PMMA composite.

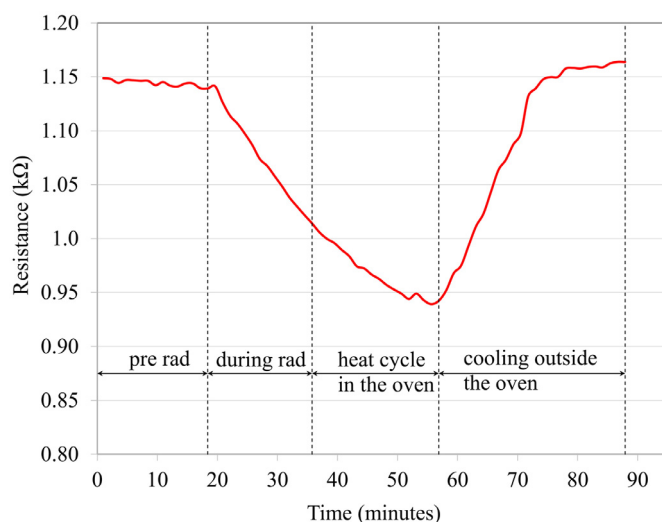


Fig. 5. Ferrocene doped MWCNT/PMMA composite heat treatment recovery of sample nominal resistance.

4. Heat treatment and reset

The radiation exposed thin film composites exhibited slow recovery times, on the order of one week. Previous work on CNT based gas sensors have shown that sensor recovery can be accelerated when subjected to heat treatment [4]. These devices recovered to pre-exposure levels after heat treatment in approximately 15 min. In this work, radiation exposed ferrocene doped MWCNT/PMMA thin film composite devices were heated to 175 °C for about 20 min and recovered to pre-radiation levels after heat treatment in approximately 15 min. Fig. 5 shows example of a heat treatment based recovery data set for a ferrocene doped MWCNT/PMMA thin film composite exposed to a radiation dose of 100 krad at a dose rate of 5.92 krad/min. Similar recovery responses were observed for the Carboxylic acid, copper coated, amine and hydrogen peroxide functionalized MWCNT/PMMA nanocomposite films.

5. Discussion

The response to the x-ray exposure of the nanocomposites in which a reduction of the thin film resistance is observed depends on the x-rays generating excess charge carriers in the films through ionization and transport of those carriers to the gold electrodes of the IDEs. For the nanocomposites containing unfunctionalized MWNTs reported in Ref. [17], ionization occurred in the MWNTs and in the PMMA matrix of the nanocomposite. For the nanocomposites containing modified or functionalized MWNT described in this work, the addition of the functional group or modifying materials may provide an additional source of excess charge carriers that are responsible for the enhanced response of some of the nanocomposites as summarized in Fig. 3. The following discussion details the possible mechanisms involved associated with the MWNT network within the nanocomposite, the PMMA matrix, and the additional material for functionalization or modification.

The reduction in resistance of the functionalized/modified MWCNT/PMMA thin film nanocomposites upon radiation exposure and the post radiation prolonged recovery observed (Figs. 2 and 5) is likely due to x-ray ionization effect [23] and subsequent charge transport of the radiation induced electron – hole pairs within the thin film nanocomposite. The transport through the film is facilitated by the network of MWCNT that permeate the composites. The importance of the MWCNT network was previously demonstrated with measurements of similar composite films composed of carbon black/PMMA in which no significant response to the x-ray exposure was observed [17].

From the dose rate experiment results summarized in Fig. 2, three distinct effects are observed. First, the resistance of the thin film composite decreases upon radiation exposure. Second, a higher radiation dose rate results in a higher rate of change in the resistance of the device. Third, the magnitude of reduction in device resistance is inversely proportional to radiation dose rate. X-ray interaction with MWCNTs in the thin film causes ionization which results in the removal of electrons from the carbon atom in the MWCNT. The removal of electrons results in sustained electron-hole pair production during radiation exposure. The increase in the number of charge carriers upon radiation exposure causes a decrease in resistance of the thin film composite, with electrons are expected to contribute the most to the conductivity increase. Radiation induced ionization in the PMMA may also contribute to the increase in conductivity of the thin film. The accelerated rate of change of resistance to increasing radiation dose rate can be explained by the enhanced electron-hole pair production due to increasing radiation intensity (photon flux) resulting in a larger x-ray ionization effect. However, this enhanced electron hole pair production due to a higher dose rate also results in an enhancement of the rate of creating defect sites [24,25] along the side walls of the nanotubes. The electric current in MWCNT primarily passes through the outermost nanotube shells [26,27], which are also the ones sustaining the most radiation induced damage. Defect sites cause reduction in electron mobility which results in reducing overall conductivity. The inverse proportionality relationship of conductivity to radiation dose rate observed in Fig. 2 is perhaps due to increased production of defect sites [17] whose conductivity reducing effects cannot be overcome by enhanced electron-hole pair production from increasing radiation intensity. In addition, the ionization of the PMMA should leave behind positively charged defects that will also induce charge carrier scattering within the films, also reducing conductivity. It should be noted that the damage site effects on the ferrocene doped MWCNT/PMMA thin film nanocomposites are significantly reversible, as evidenced by the repeatable responses shown in Fig. 4, and the accelerated heat treatment recovery process discussed in Section 4.

The MWCNT network in the thin films facilitates the separation of the charge carriers, electron-hole pairs [28,29]. Some of these holes, which have relatively low mobility, will likely be trapped within the oxide of the IDE and the polymer matrix of the MWCNT/PMMA film, which may explain the long recovery time in the nominal resistance of the thin film devices. Thermal treatment (described in Section 4) of the irradiated composite provides sufficient thermal energy for the holes to escape, thereby relaxing the composite in much shorter time compared to annealing at room temperature.

Radiation results summarized in Fig. 3 demonstrates the overall higher sensitivity of functionalized/modified MWCNT/PMMA composite thin films to x-ray radiation over unfunctionalized [17] MWCNT/PMMA nanocomposite thin films. The increased sensitivity of functionalized MWCNT/PMMA nanocomposites can be attributed to the presence of functional groups on the MWCNT which are not present on the unfunctionalized MWCNT. It has been shown previously that conductivity can be altered by the presence of functional groups on the outer most wall, though the conductivity has been preserved in the inner tubes of double-walled CNT [30]. These functional groups provide an additional source of excess charge carriers from broken bonds during x-ray interactions thereby providing more charge carriers for conduction compared to the unfunctionalized MWCNT, hence increasing the sensitivity. Among all the functionalized groups, the high sensitivity of ferrocene doped MWCNT/PMMA composite is likely due to the presence of extra ferrocene (an electron donor [31]) groups on PMMA which are not present on the unfunctionalized MWCNT/PMMA composite. The electron donating ferrocene groups break bonds during x-ray interaction thereby providing more charge carriers for conduction, hence increasing the sensitivity compared to the other nanocomposites. For the COOH MWCNT, the COOH functional groups on the MWCNT would serve as a source of negative charge carriers and has

been shown to increase electron transfer rate due to the presence of COOH functional groups [32]. In the case of the copper modified MWCNT, the copper may provide conductive paths through the junctions between the nanotubes, and as discussed earlier, dispersion of the functionalized and modified MWCNTs in the PMMA also contribute to the ability of the nanotube network to be conductive.

For the hydrogen peroxide and amine functionalization groups, mechanisms providing extra charge carriers would be smaller in magnitude compared to the COOH and the copper coated MWCNT, giving a smaller overall resistance change during irradiation. The harsh acidic process of making the hydrogen peroxide MWCNT would damage the sidewall and result in reduced conductivity compared to pristine MWCNT due to increased carrier scattering. In obtaining the amine functionalization, the COOH MWCNT are further derivatized with a long chain aliphatic amino group to produce the amine MWCNT which impedes charge carriers at the sidewall, and thus reducing charge transfer to the inner tubes of the MWCNT [33]. The reduction of charge carriers at the sidewall decreases the sensitivity of the amine functionalized nanotubes. The role of different functionalized groups on physical properties has been shown previously [34].

As suggested above, an alternative to the scenario described in which the primary source of excess carriers results from ionization effects on the side walls of the MWCNTs, the ionization effects generating excess electron-hole pairs may reside primarily in the PMMA matrix. The x-rays may produce ionization tracks within the PMMA matrix that may then help transport the charge carriers through the film to the conductive MWCNT network within the films. Also, the film composition is about 96.6% PMMA which would favor the PMMA as a principal source of excess charge carriers due to x-ray ionization. It is possible that a combination of ionization in the MWCNT and the PMMA contribute to the observed results. However, prior results [35] indicate that ionization of SWCNT from x-rays does take place, as there was no polymer matrix in the samples used in Ref. 35. Further modeling and experimental study may help elucidate the responsible mechanism(s).

6. Conclusion

It has been demonstrated in this study that ferrocene doped MWCNT/PMMA thin film nanocomposites exhibited a repeatable accelerated change in the reduction of resistance to increased radiation dose rates. Ferrocene doped MWCNT/PMMA composite thin film showed increased sensitivity to x-ray radiation compared to unfunctionalized MWCNT/PMMA composite [17] and all other functional groups used in this study. The thin film composites in this study had slow recovery post-radiation which was accelerated by heat treatment. These results show promise in using ferrocene doped MWCNT/PMMA based thin film composites for ionizing radiation detection and dosimetry.

Conflict of interest

None.

Acknowledgments

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Supplementary materials

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